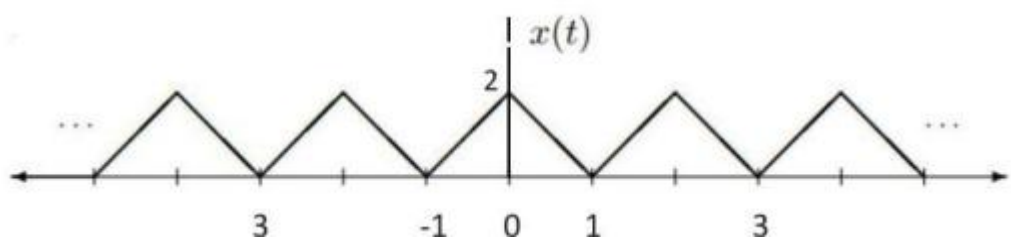
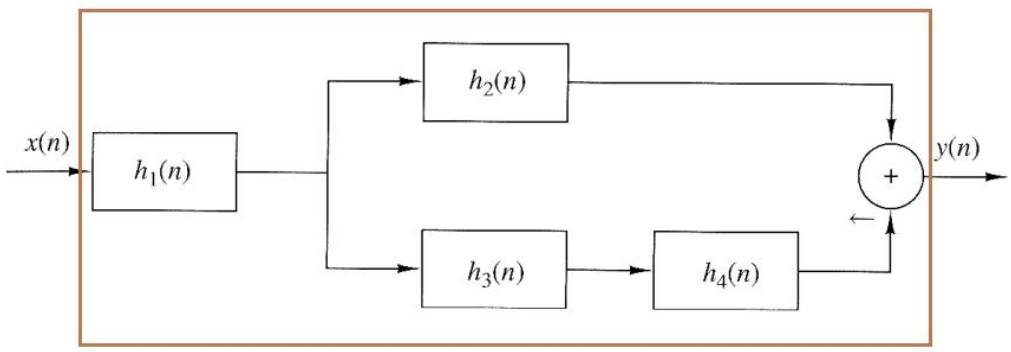
Subject: Signals & Systems (EC 20001) (Question Bank)

Question Number	Questions
Q1.	Answer the following question in short.
a.	What are the even and odd components of $u(t)$?
b.	A continuous time signal $x(t)$ is first left shifted by 5 then reversed in time and then amplitude scaled by a factor of 6. Express the resultant signal $y(t)$ in terms of $x(t)$.
c.	Mention the Dirichlet's condition for the existence of Fourier Series
d.	Explain the periodicity property of the DTFT.
e.	Find the period of the following signal:$x(n) = 5 + \cos\left(\frac{3\pi}{4}n\right) + 2\sin\left(\frac{5\pi}{4}n\right)$
f	Find the fundamental period of the signal (a). $x(t) = je^{jt}$ (b). $x(t) = e^{-(1+j)t}$
g.	Define even and odd signals with relevant examples.
h.	Evaluate the Fourier transform of a unit step function $u(t)$.
i.	If the Fourier Transform of $x(t)$ is $\left(\frac{2}{w}\right)\sin(\pi w)$, then what is the FT of $e^{j5t}x(t)$.
j.	By utilizing the property of FT, evaluate the FT of $x(t) = e^{at}u(-t), a > 0$.
k.	How to determine the correlation of two signals using convolution.
Q2.	(a) A causal discrete time signal is given by $x(n) = \{2, 1, 3, 4, 1, 2\}$. Plot the signal $x(-0.5n + 3)$. Find the energy and power of $x(n)$.
Q3.	Classify the following continuous time system into causal/non causal, linear/non-linear and time variant/time invariant: $y(n) = n\,x(n)$
Q4.	Determine the output of a causal system having impulse response of {3,3,2,2} for

	a causal input of {2, 1, 3} using graphical method of convolution. Verify your answer using tabular method of convolution.
Q5.	Determine the cross correlations $R_{xy}(n)$ and $R_{yx}(n)$ of the following signals: $x(n) = \{2,2,3\}$, $y(n) = \{2,4,1,4\}$. Is there any relation between the two correlations? Find the autocorrelation of $x(n)$.
Q6.	Find the DTFT of the following signals (a). $x(n) = a^n u(n), \lvert a\rvert < 1$ (b). $x(n) = a^n u(-n-1), \lvert a\rvert > 1$
Q7.	Find the inverse DTFT of the following signals (a). $X(w) = \delta(w)$ (b). $X(w) = e^{-jw}[1 + \cos(w)]$
Q8.	Find the Trigonometric and the Exponential Fourier series coefficients for the waveform given below 
Q9.	Check whether the signal $x(t) = e^{-a\lvert t\rvert}$ is an energy signal or power signal.
Q10.	Evaluate the Fourier Transform of the following signals (a). $x(t) = \delta(t-3)$. Also plot its response in time and frequency domain. (b) $x(t) = sgn(t)$. Also plot its magnitude and phase response
Q11.	Given $x(n) = u(n)$ and $h(n) = 2^{-n}u(n-1)$. Find the convolution sum of the sequences.
Q12.	(a). State the commutative property of convolution (b). Determine the output $y(n)$ of a system where the input $x(n) = u(n+1) -$

	$u(n-5)$ and the system impulse response $h(n) = (1,1,1,1)$ using graphical method.
Q13.	$x(n) = (1, \mathbf{0}, 2,3,4,1)$. (1) For the given signal, determine the even component and odd component. (2). Determine the energy and power of $x(n)$.
Q14.	Use deconvolution method to determine the impulse response $h(n)$ of the system given the output $y(n) = \left\{4 \quad 13 \quad \underset{\uparrow}{28} \quad 27 \quad 18\right\}$ and the input is $x(n) = \left\{1 \quad \underset{\uparrow}{2} \quad 3\right\}$.
Q15.	 $h(n)$ Consider the interconnected LTI system as shown in figure. Determine overall $h(n)$, if $h_1(n) = \left\{1 \quad \underset{\uparrow}{2} \quad 1\right\}$, $h_2(n) = \left\{3 \quad \underset{\uparrow}{2} \quad 1\right\}$, $h_3(n) = \{1 \quad 1\}$, $h_4(n) = \{2 \quad 2\}$ are given respectively.
Q16.	Given signal $x(n) = \frac{1}{3}\{\delta(n+1) + \delta(n) + \delta(n-1)\}$. Plot its magnitude response and phase response. Also determine value $X(0)$.
Q17.	